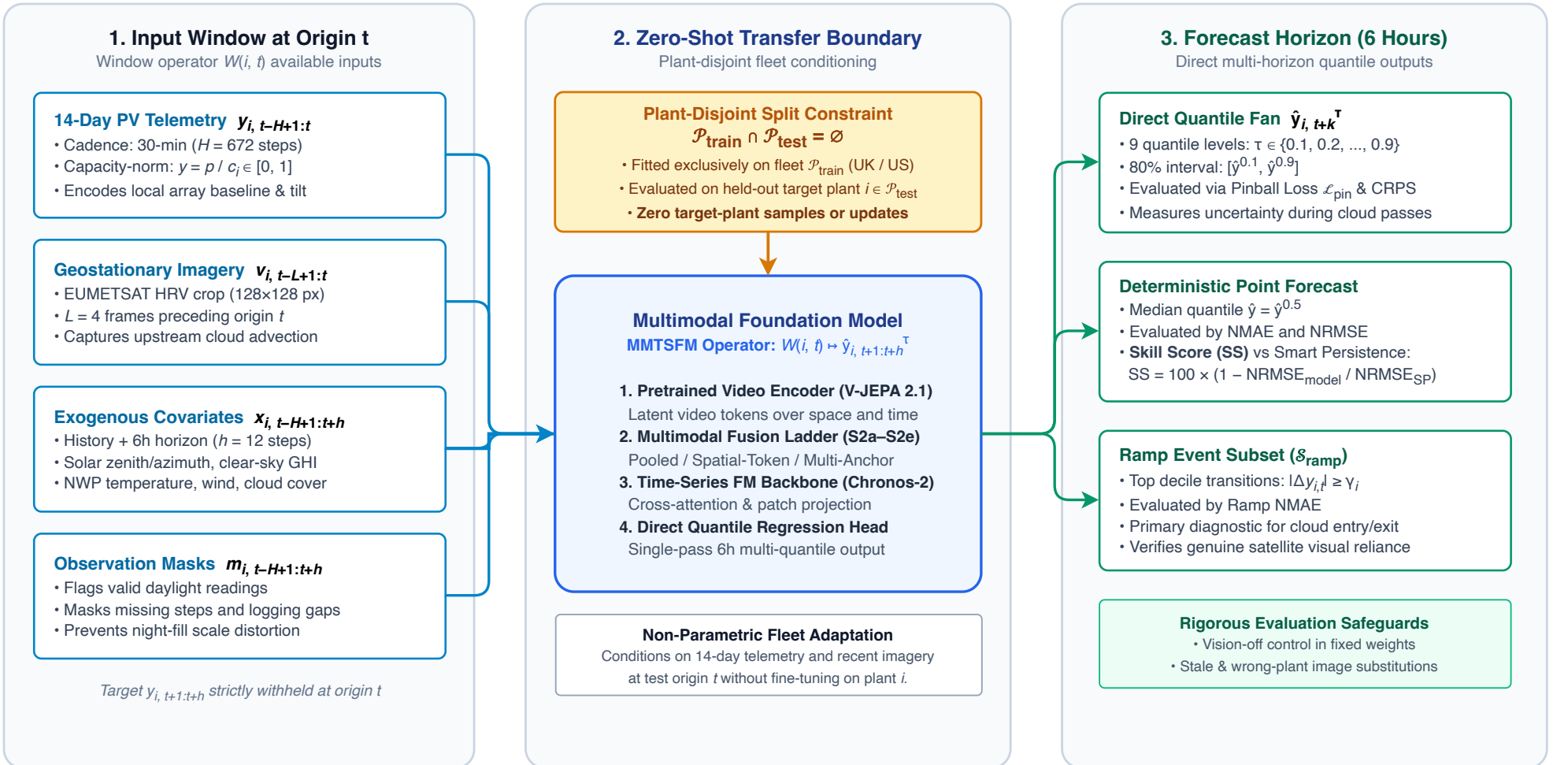

1. Input Window at Origin t
Window operator W(i, t) available inputs
14-Day PV Telemetry y_{i, t−H+1:t}
• Cadence: 30-min (H = 672 steps)
• Capacity-norm: y = p / c_i ∈ [0, 1]
• Encodes local array baseline & tilt
Geostationary Imagery v_{i, t−L+1:t}
• EUMETSAT HRV crop (128×128 px)
• L = 4 frames preceding origin t
• Captures upstream cloud advection
Exogenous Covariates x_{i, t−H+1:t+h}
• History + 6h horizon (h = 12 steps)
• Solar zenith/azimuth, clear-sky GHI
• NWP temperature, wind, cloud cover
Observation Masks m_{i, t−H+1:t+h}
• Flags valid daylight readings
• Masks missing steps and logging gaps
• Prevents night-fill scale distortion
Target y_{i, t+1:t+h} strictly withheld at origin t
2. Zero-Shot Transfer Boundary
Plant-disjoint fleet conditioning
Plant-Disjoint Split Constraint
P_train ∩ P_test = ∅
• Fitted exclusively on fleet P_train (UK / US)
• Evaluated on held-out target plant i ∈ P_test
• Zero target-plant samples or updates
Multimodal Foundation Model
MMTSFM Operator: W(i, t) ↦ ŷ_{i, t+1:t+h}^τ
1. Pretrained Video Encoder (V-JEPA 2.1)
Latent video tokens over space and time
2. Multimodal Fusion Ladder (S2a–S2e)
Pooled / Spatial-Token / Multi-Anchor
3. Time-Series FM Backbone (Chronos-2)
Cross-attention & patch projection
4. Direct Quantile Regression Head
Single-pass 6h multi-quantile output
Non-Parametric Fleet Adaptation
Conditions on 14-day telemetry and recent imagery at test origin t without fine-tuning on plant i.
3. Forecast Horizon (6 Hours)
Direct multi-horizon quantile outputs
Direct Quantile Fan ŷ_{i, t+k}^τ
• 9 quantile levels: τ ∈ {0.1, 0.2, ..., 0.9}
• 80% interval: [ŷ^0.1, ŷ^0.9]
• Evaluated via Pinball Loss ℒ_pin & CRPS
• Measures uncertainty during cloud passes
Deterministic Point Forecast
• Median quantile ŷ = ŷ^0.5
• Evaluated by NMAE and NRMSE
• Skill Score (SS) vs Smart Persistence:
SS = 100 × (1 − NRMSE_model / NRMSE_SP)
Ramp Event Subset (𝒮_ramp)
• Top decile transitions: |Δy_{i,t}| ≥ γ_i
• Evaluated by Ramp NMAE
• Primary diagnostic for cloud entry/exit
• Verifies genuine satellite visual reliance
Rigorous Evaluation Safeguards
• Vision-off control in fixed weights
• Stale & wrong-plant image substitutions